

# Four Months' Notice

## Turning El Niño into a dengue early-warning alarm for ASEAN

A freely available climate index gives ASEAN health offices months of warning before a dengue season turns severe. We turn that lag into a two-condition alert rule that any provincial health office can run today, with no model, no code and no paid data.

SDG 3 Health

SDG 13 Climate

SDG 6 Water

4

months of public notice before  
a dengue season turns severe

Team Swallowgank | Institut Teknologi Sepuluh Nopember | Indonesia

## THE PROBLEM

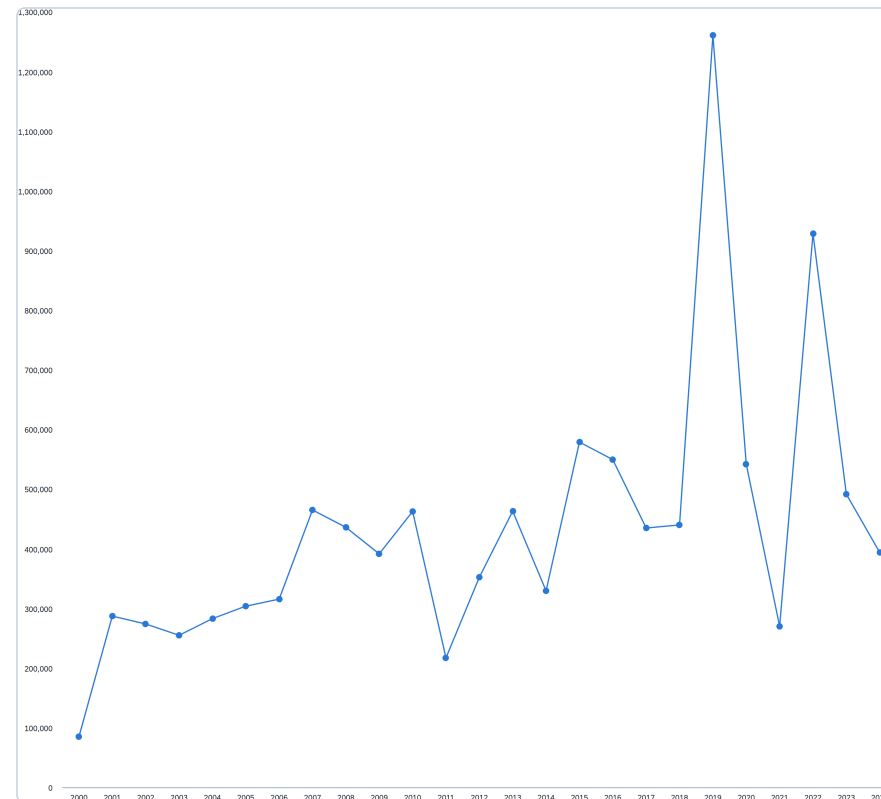
# Dengue is ASEAN's fastest-moving climate-sensitive disease

## 1.26M

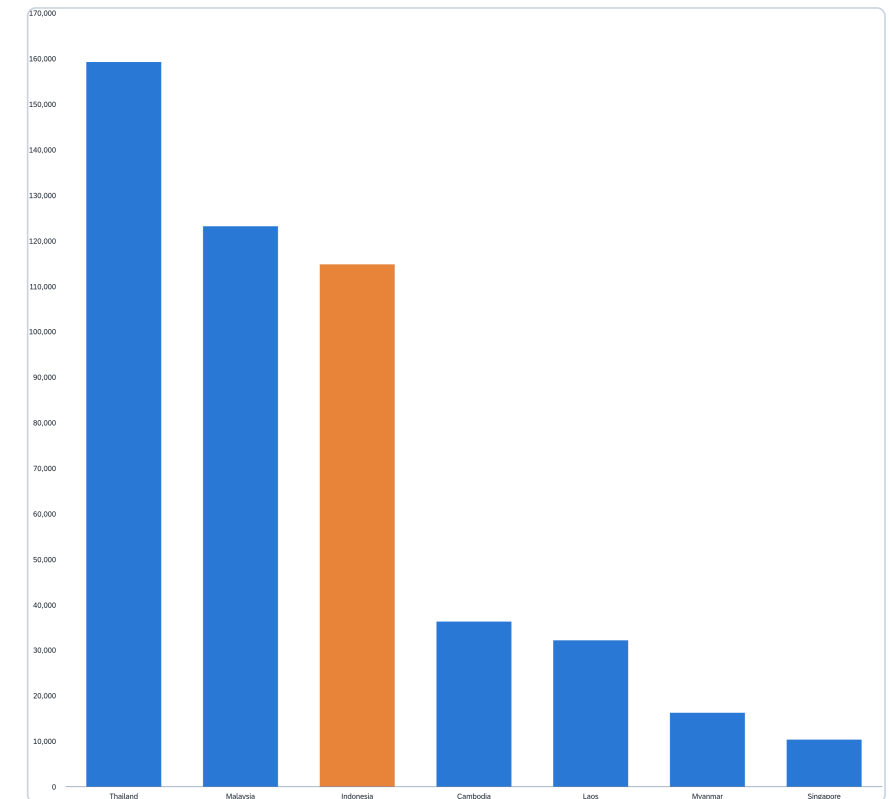
cases in 2019, the worst year in our nine-country panel

- Vietnam 368k, Philippines 220k, Thailand 159k (latest complete year each)
- The 2023–24 El Niño added an estimated **9.6 million cases worldwide** (Tian et al., 2025)
- Outbreaks are still fought **reactively**, only once cases surge

### ASEAN dengue cases, 2000–2024



### Cases by country, latest complete year



The burden is large, rising and spiky. Spikes are what an alarm can catch; a trend line is not something a health office can act on.

## THE QUESTION

# Climate data arrives months before patients do

Can a freely available climate index tell a health office how bad the coming season will be, early enough to act?

- **SDG target 3.3:** end the epidemics of neglected tropical diseases, of which dengue is one
- **SDG target 13.1:** strengthen resilience and adaptive capacity to climate-related hazards
- **SDG target 6.1:** safe water for all. Where piped water is absent, households store it, and *Aedes* breeds in the store

From public index to public action



**ASEAN priority.** Dengue sits under **ASEAN Health Cluster 2, Responding to All Hazards and Emerging Threats**, whose ASEAN Dengue Day statements ask member states for earlier and better coordinated action.

## WHAT WOULD MAKE US WRONG

Signal is only the wet season

[answered p. 7](#)

Rule was tuned on its own test years

[answered p. 12](#)

Our innovation angle does not hold

[answered p. 14](#)

## DATA &amp; METHOD

# Nine open datasets, one monthly panel, zero paid sources

| DATA                                | ROWS   | COVERAGE                          |
|-------------------------------------|--------|-----------------------------------|
| Dengue cases (OpenDengue V1.3)      | 70,397 | 10 countries, 1955–2025           |
| Dengue × climate monthly panel      | 55,329 | province-month, lags pre-computed |
| Temperature & rainfall (CCKP, ERA5) | 9,000  | monthly, 1950–2024                |
| ONI El Niño index (NOAA CPC)        | 918    | monthly, 1950–2026                |
| Water & sanitation (WHO/UNICEF JMP) | 8,692  | 2000–2024                         |
| Population (BPS Indonesia)          | 140    | provinces, denominators           |
| Flood exposure (GHSL)               | 30,348 | 836 cities, 1975–2030             |

175k

rows in the joined panel

9

countries with monthly data

212

provinces covered

0

paid or login-only sources

SDG 3

SDG 13

SDG 6

Every table joins on the same two keys, `iso3` and `year`, with `period` added for the monthly dengue × climate panel. Nothing is entered by hand.

## METHOD

- Monthly national series, 2010+
- Lagged correlations, 0–4 months
- **Anomalies:** the seasonal cycle is removed before we accept any correlation
- A two-condition trigger rule
- Out-of-sample test on unseen years
- WHO/PAHO endemic-channel benchmark

All charts built in SAP Analytics Cloud.

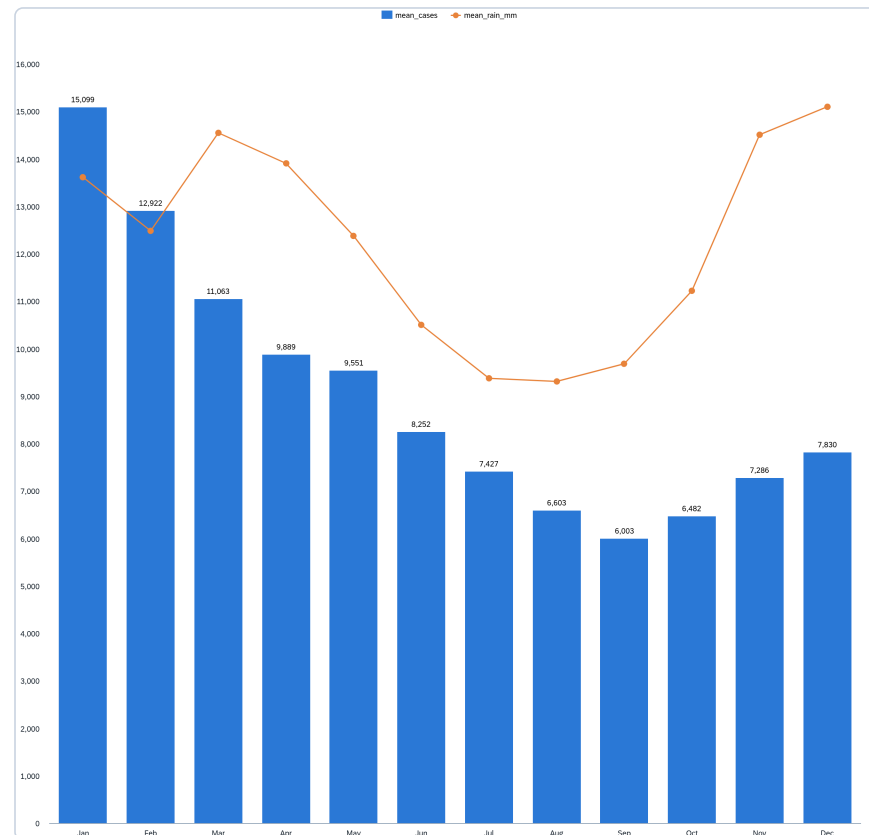
## ANALYSIS 1 · THE SEASON

# Dengue has a calendar, and it is not the same one everywhere

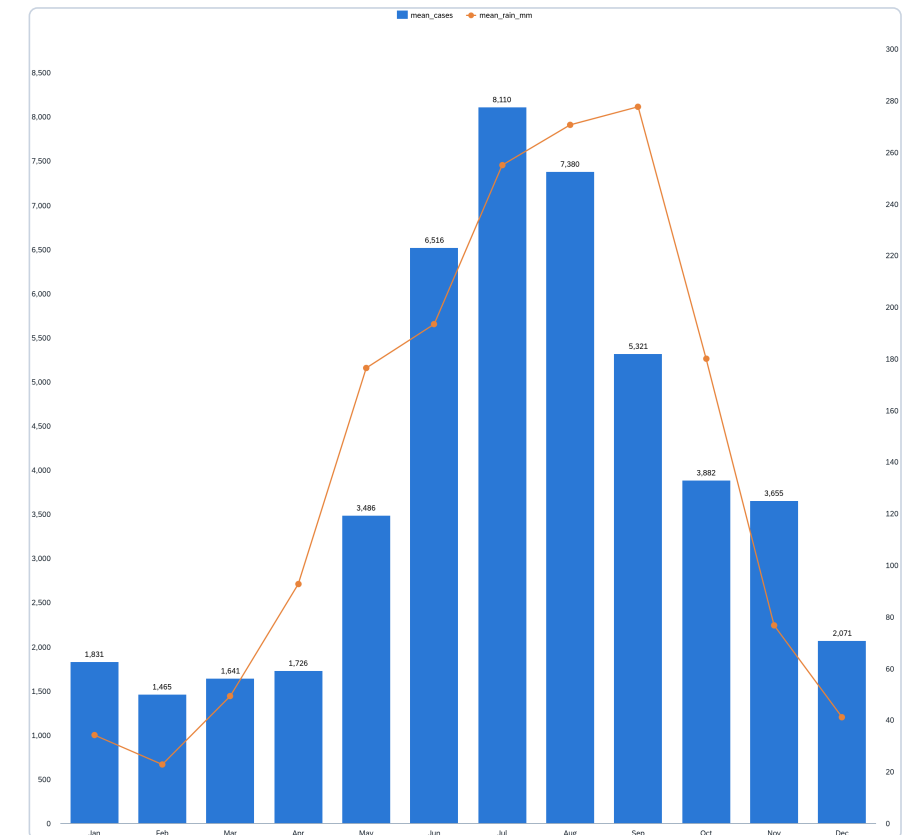
- **Indonesia** peaks **Jan–Apr**, right after the December rainfall peak  
Jan 15,099 cases/month against Sep 6,003 = 2.5×
- **Thailand** climbs with the **onset** of the wet season and peaks **Jun–Sep**  
Jul 8,110 against Feb 1,465 = 5.5×
- One disease, two calendars in opposite halves of the year

A regional alarm has to be local in its timing.

Indonesia: cases vs rainfall by month



Thailand: cases vs rainfall by month



Rainfall alone cannot set the date: Thailand's rain peaks in September, two months *after* its cases. Something earlier in the chain is doing the work.

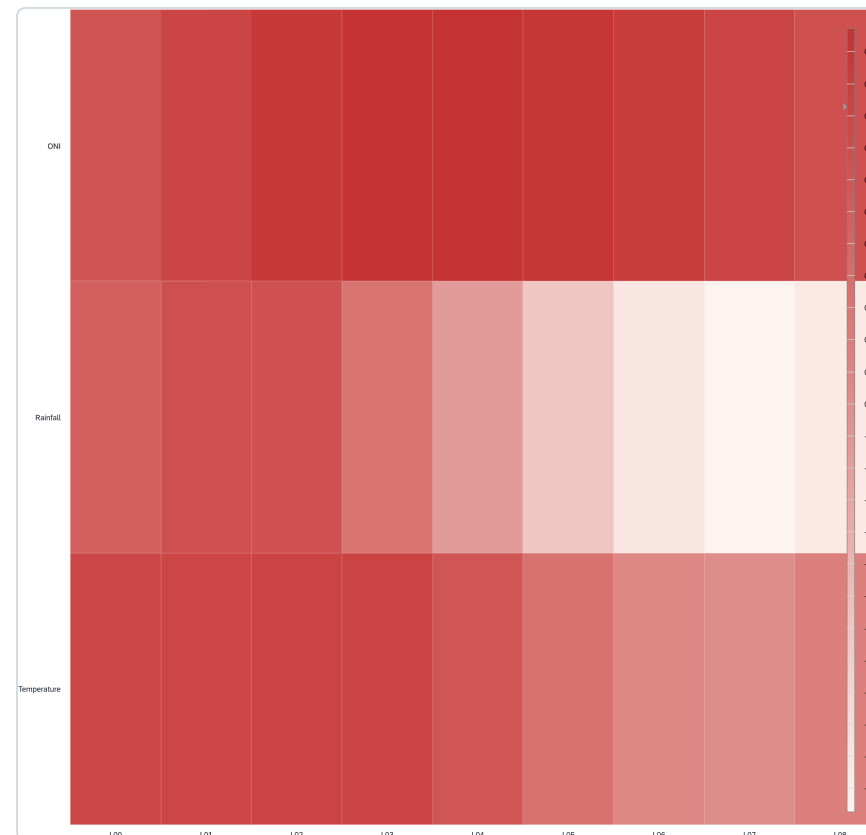
## ANALYSIS 2 · THE LAG

# The climate signal arrives 1–4 months before the cases

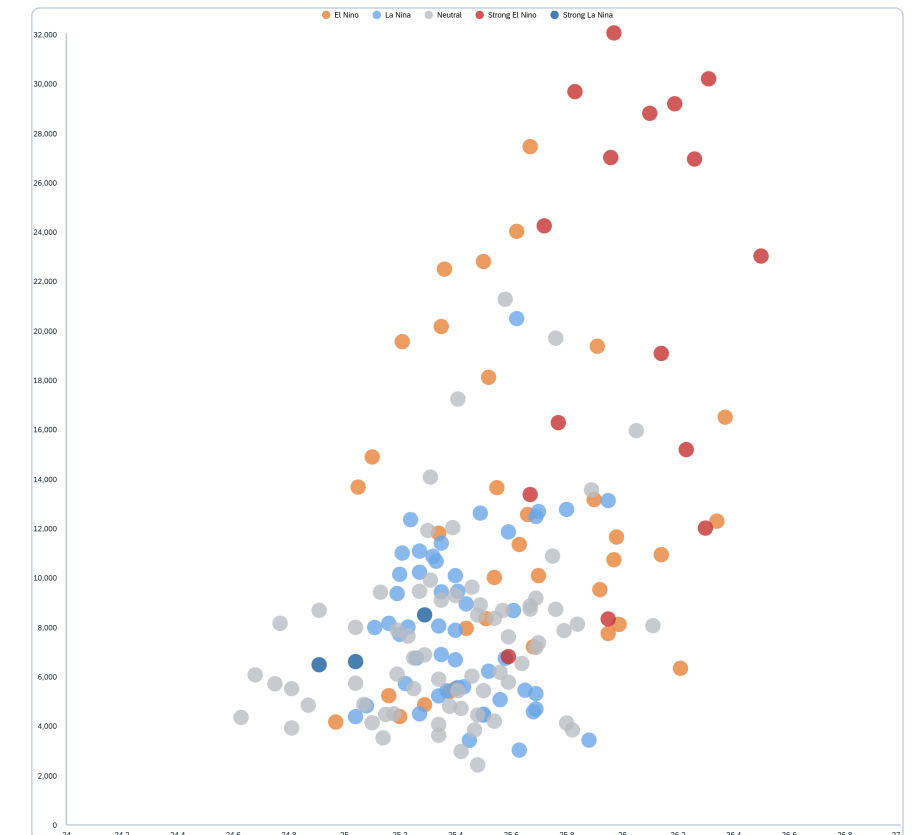
| PREDICTOR        | LAG     | R     |
|------------------|---------|-------|
| <b>INDONESIA</b> |         |       |
| ONI (El Niño)    | 4 mo    | +0.59 |
| Temperature      | 3 mo    | +0.47 |
| Rainfall         | 1 mo    | +0.40 |
| <b>THAILAND</b>  |         |       |
| Temperature      | 3 mo    | +0.57 |
| Rainfall         | same mo | +0.48 |

The lag *is* the opportunity:  
it is the time a health of-  
fice has to act.

Correlation by predictor and lag



Temperature (lag 2) vs cases, Indonesia



Every predictor peaks at a lag of one month or more, so none of them is a coincident symptom. The best one, ONI, is also the one published earliest and freely.

## ANALYSIS 3 · THE STRONGEST SIGNAL

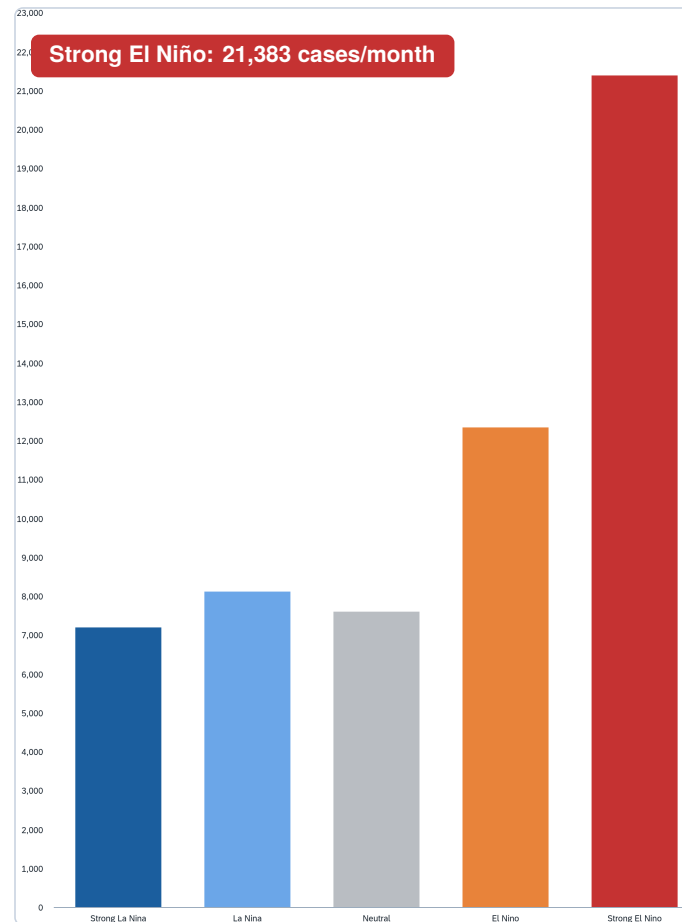
# A strong El Niño nearly triples Indonesia's monthly dengue burden

## 2.8×

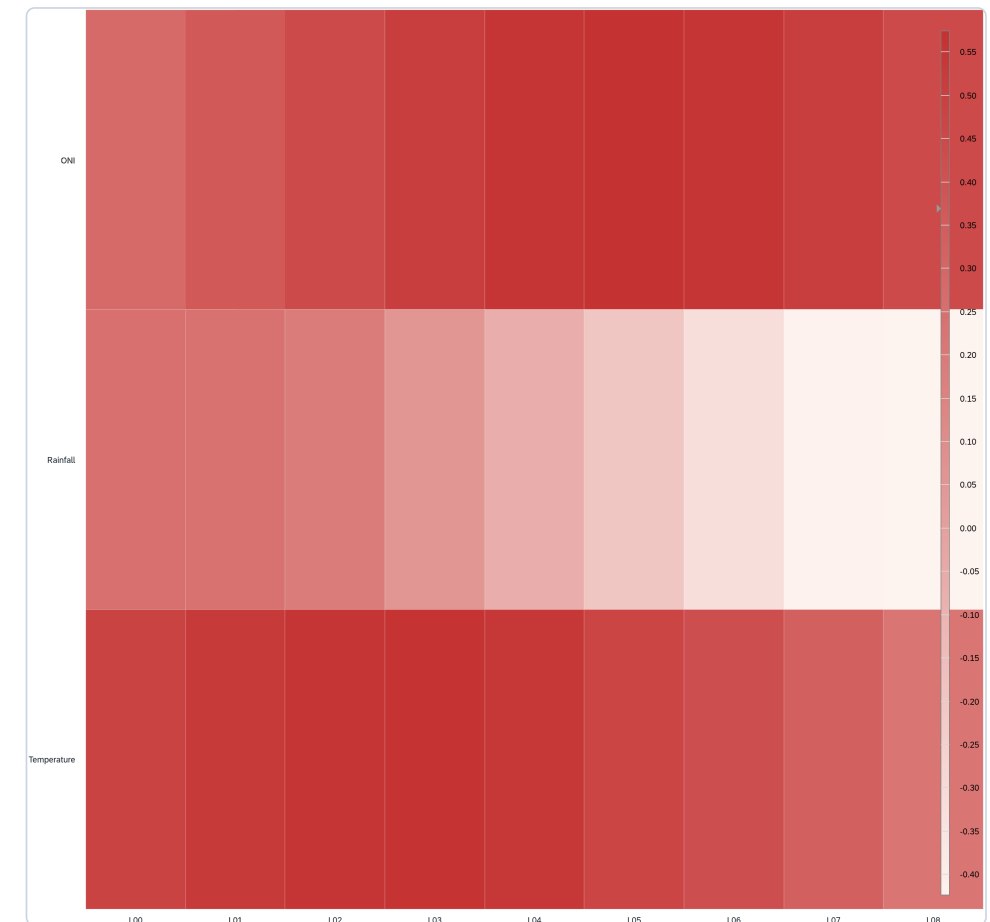
21,383 vs 7,597 cases per month,  
grouped by the ENSO phase four  
months earlier

- The grouping uses the ENSO phase **four months earlier**, so this is a forecast rather than hindsight
- **Not merely seasonality.** With the seasonal cycle removed, the ONI correlation still holds at  $r = +0.56$
- **Not an aggregation artefact.** Cases run higher in El Niño months in **33 of 34 Indonesian provinces** (sign test  $p = 2 \times 10^{-9}$ ) and in 70 of 77 Thai provinces
- Rainfall does *not* survive the same test, falling from +0.40 to +0.24, and we report that

Mean monthly cases by ENSO phase (lag 4)



Same signal, seasonality removed



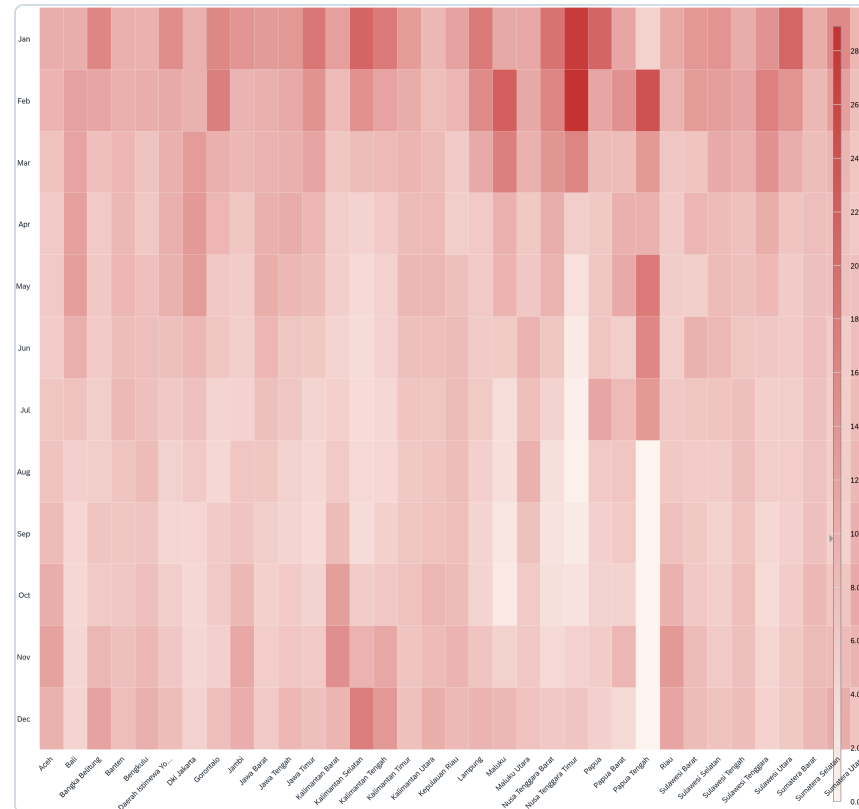
## ANALYSIS 4 · WHERE AND WHEN

# The alarm must be set province by province

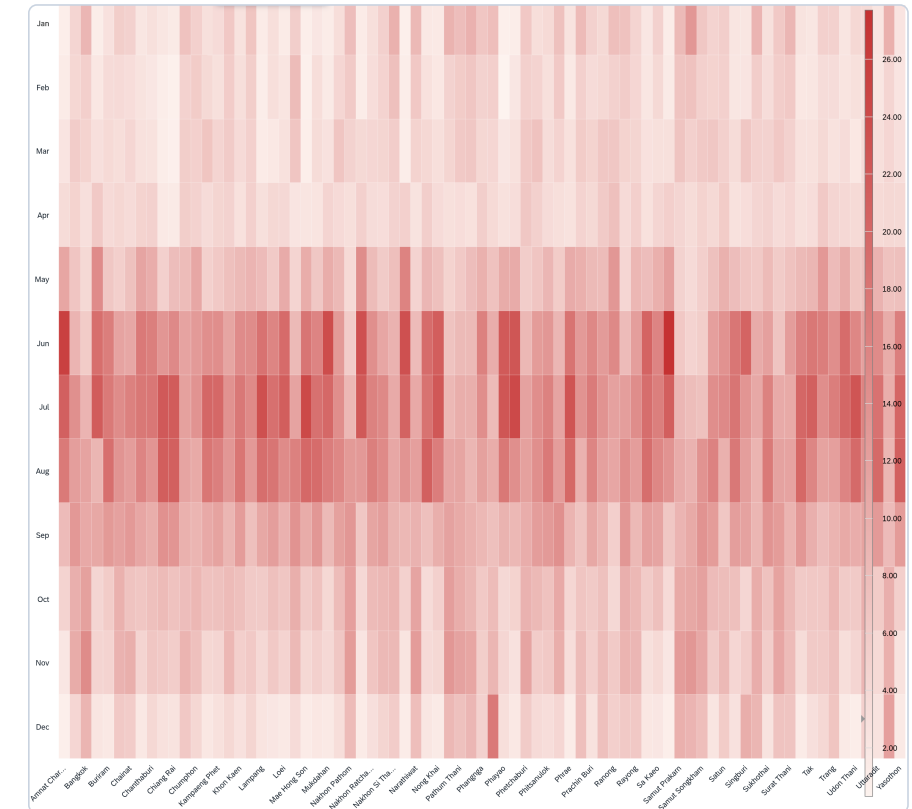
- **Indonesia:** 20 of 35 provinces peak in **January**, 8 in February, but **Bali peaks in May** and North Kalimantan in December
- **Thailand:** 38 of 77 provinces peak in **July**, 16 in August
- A single national alert date would be wrong for a third of Indonesia

**Same colour scale, same twelve rows. Only the calendar moves.**

### Indonesia: share of annual cases by month



### Thailand: share of annual cases by month



This is why the rule carries a month condition as well as a climate one. The index says how bad; the province's own calendar says when.

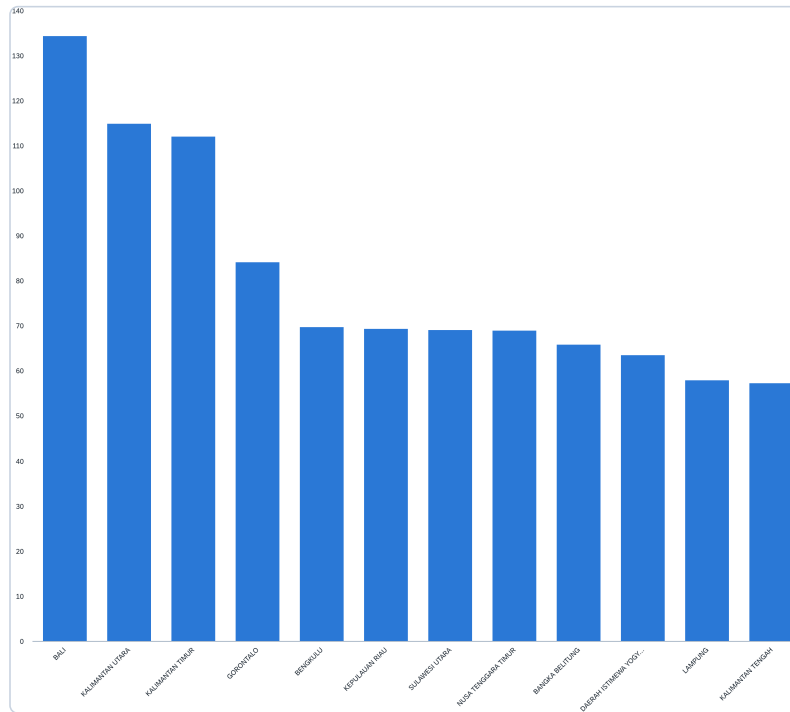
## ANALYSIS 5 · WHO IS HIT HARDEST

# The same alarm, aimed at the most exposed places

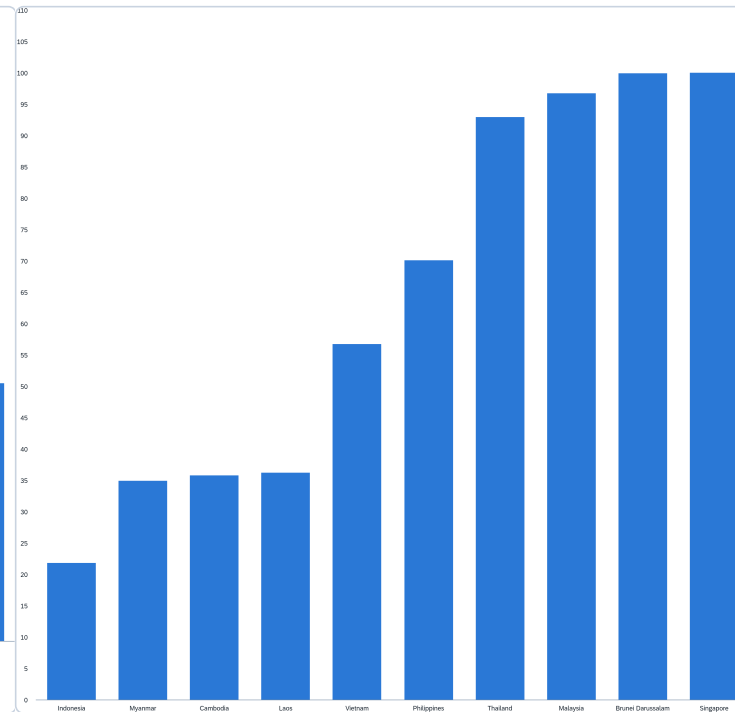
- Highest incidence 2018–2020: **Bali 134**, North Kalimantan 115, East Kalimantan 112 per 100k, which is **14× the lowest province** (Papua, 9)
- Indonesia has the **lowest piped-water access in ASEAN** at 21.8% (JMP 2024). Households therefore store water, and stored water breeds *Aedes*

- **12.9 million people in Bangkok** and 1.7 million in Jakarta live in the 1-in-100-year flood zone
- We use these three layers to **target** the alarm, not to explain the disease. The one place we tried to explain it, the claim failed (p. 14)

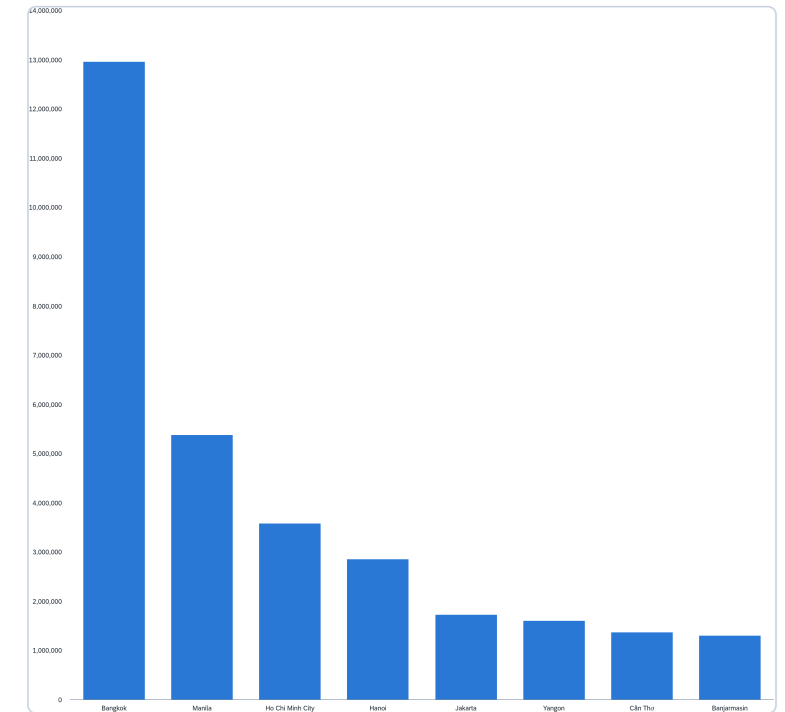
Incidence per 100k, 2018–2020



Piped water access, ASEAN



Flood-zone population, top cities



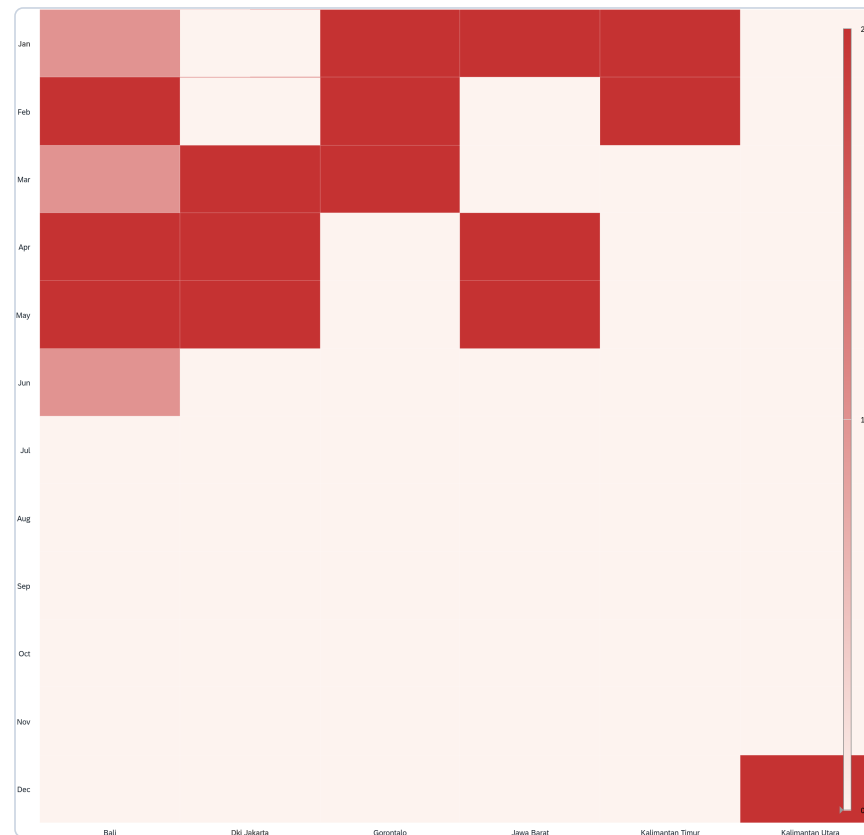
## SYNTHESIS

# From signal to schedule: a risk calendar every health office can read

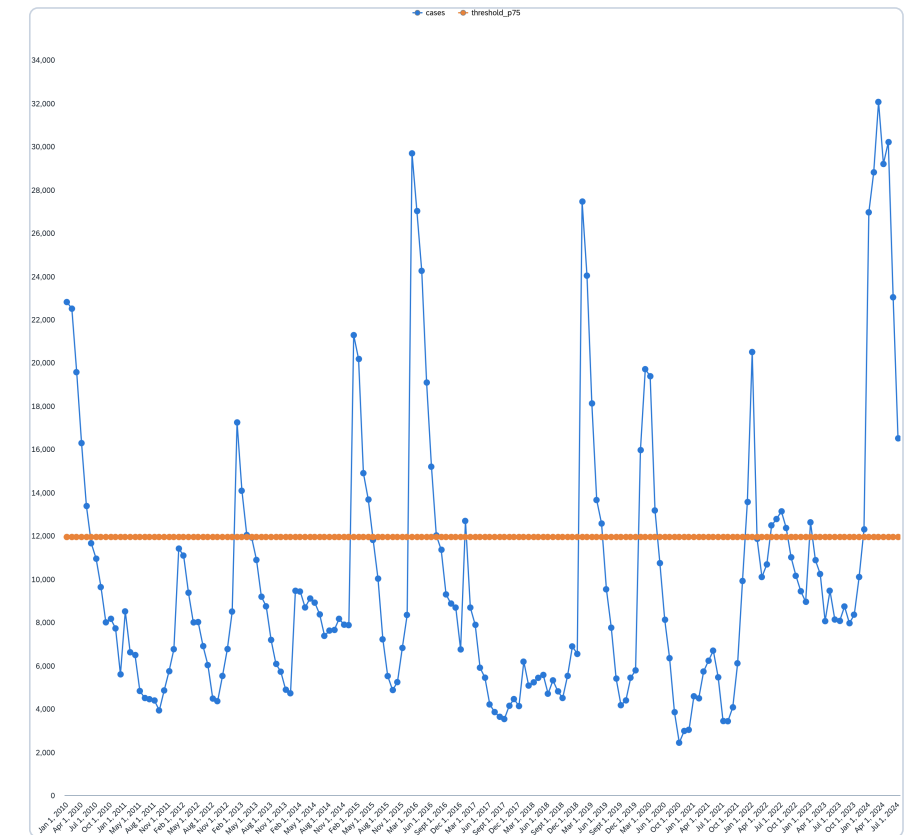
- Each province gets three states, **Peak**, **Watch** and **Normal**, derived from its own case history
- The El Niño index sets **how severe** the coming peak will be; the calendar sets **when** it arrives
- Real lead times, every El Niño episode since 2009: **0–3 months** on observed ONI, longer still on NOAA and BMKG forecast ONI

Two public numbers, one page, no model to maintain.

Risk calendar, six priority provinces



Cases vs the P75 outbreak threshold



The threshold is the WHO-style P75 of the province's own history, not a number we picked. Everything above the orange line is what the alarm is trying to anticipate.

## THE SOLUTION

# Two conditions. No model. No code.

**“If the month is Jan–Apr AND ONI four months ago was  $\geq 0.5 \rightarrow$  raise alert”**

## 11.6%

of months fire the alert

## 100%

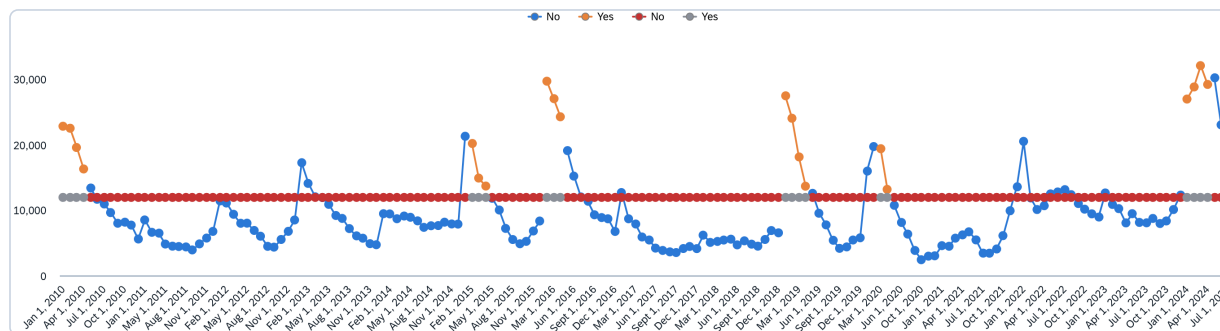
precision, every alert in a top-quartile month

## 4×

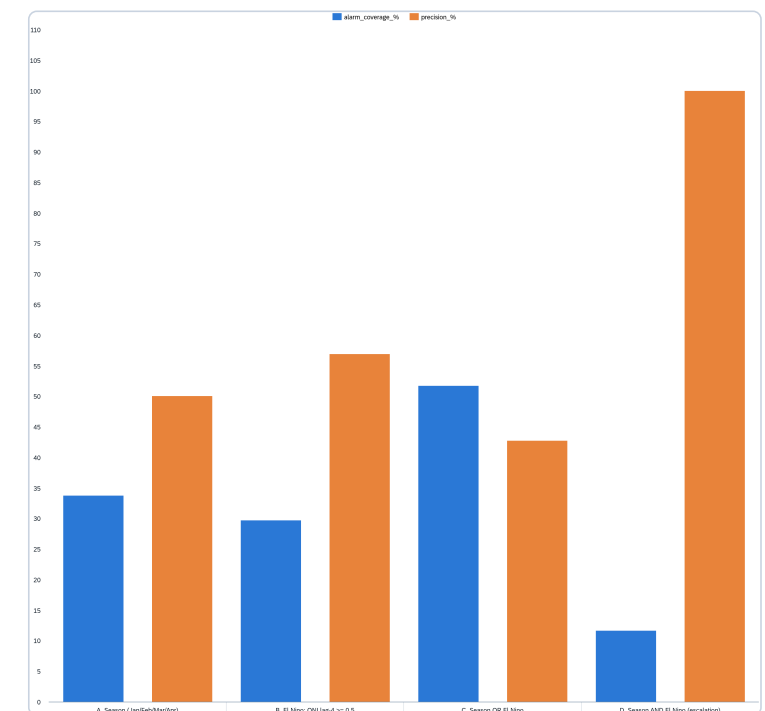
lift over the base rate

- Both inputs are public and free: NOAA publishes ONI monthly, the calendar comes from the office's own records
- It runs in a spreadsheet, a WhatsApp broadcast or the SAC page on p.15. Where a statistical forecast needs a trained model and a data team, a provincial office can start this next week

### When the rule fired, 2010–2024



### Precision vs alert frequency, four rules

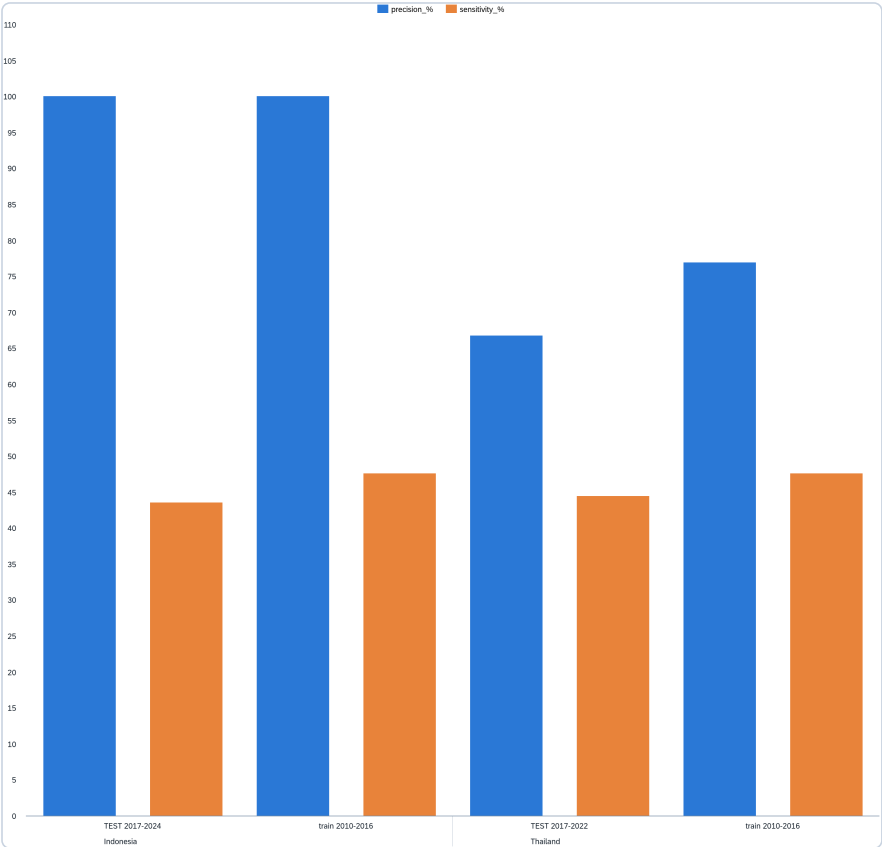


VIABILITY

We tried to break our own rule. It held.

- **Out-of-sample test:** rule built on 2010–2016, tested blind on 2017–2024 → **precision still 100%**, sensitivity 44%, lift 3.9
- Against the **WHO/PAHO endemic-channel** definition, which removes seasonality by construction, specificity is **91%** and lift 2.5. *All* of the value comes from the El Niño condition rather than from the season
- Published dengue early-warning systems report PPV 43–86% (17-system review). We reach a comparable PPV **using monthly data and no statistical model**
- Replicates in Thailand with its own best predictor (temperature anomaly, lag 2): lift 2.7

Train vs blind test, Indonesia and Thailand



How we compare to published EWS

| SYSTEM                                          | SENS. % | PPV % |
|-------------------------------------------------|---------|-------|
| 17 published EWS (range)<br>Baharom et al. 2022 | 50–100  | 43–86 |
| EWARS-csd, Colombia<br>Schlesinger et al. 2024  | 97      | 75    |
| WHO/TDR EWARS, Mexico<br>Cárdenas et al. 2021   | 100     | 83    |
| Rule D vs P75 (blind test)<br>this deck         | 44      | 100   |
| Rule D vs endemic channel<br>this deck          | 29      | 30    |

ADOPTION IN THREE PHASES

- 1 Bali, N. & E. Kalimantan pilot → 2 all 38 provinces → 3 ASEAN peers

Cost: open data, monthly refresh, existing staff. No new licence and no new headcount.

We report the endemic-channel row even though it is our worst result. It is the harder benchmark, and hiding it would be the kind of selective reporting this deck argues against.

IMPACT · INDONESIA ALONE

Acting 4–8 weeks earlier is worth US\$1.5–3.7 million a year

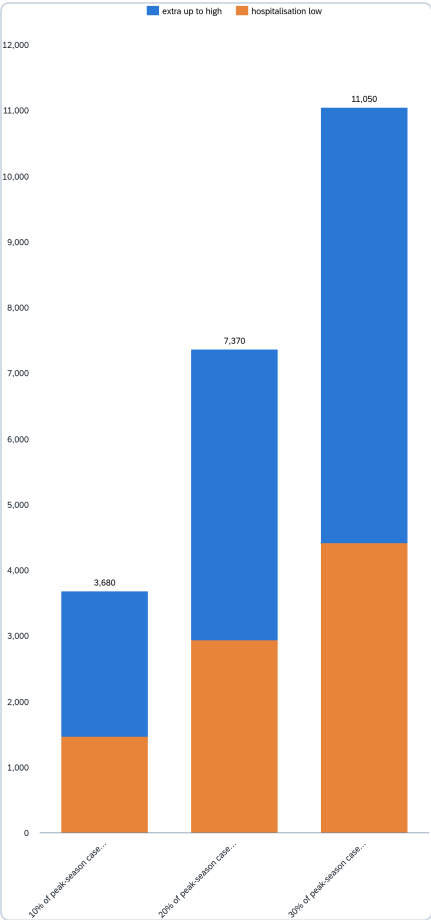
4,656

cases/year prevented in Indonesia, under the most conservative scenario

- Indonesia averages **104,753 cases/year** (complete years 2015–2023); **44% fall in Jan–Apr**
- Prevent just **10%** of peak-season cases (an assumption, not a finding) = 4,656 cases/year
- At Indonesia's published hospitalisation cost of US\$316–791/case → **US\$1.5–3.7 M (Rp 24–59 billion)/year**
- For scale, dengue costs Indonesia **US\$381–681 M/year** (Nadjib 2019; Wilastonegoro 2020)

Every figure here is either published or a stated assumption. None of them is a model output.

Avoided cost by scenario



Lead time, every El Niño episode since 2009

| EPISODE | ALARM FIRES | CASES CROSS P75 | LEAD TIME |
|---------|-------------|-----------------|-----------|
| 2009/10 | Dec 2009    | Jan 2010        | 1 month   |
| 2014/16 | Feb 2015    | Feb 2015        | 0         |
| 2018/19 | Jan 2019    | Jan 2019        | 0         |
| 2019/20 | Mar 2020    | Mar 2020        | 0         |
| 2023/24 | Sep 2023    | Dec 2023        | 3 months  |

THE LEAD TIME IS OBSERVED, NOT ASSUMED

Three of five episodes gave zero months on *observed* ONI, and we print that. The value comes from the two that did not, and from the fact that NOAA and BMKG publish a *forecast* ONI months ahead of the observed one, on which the same rule fires earlier still.

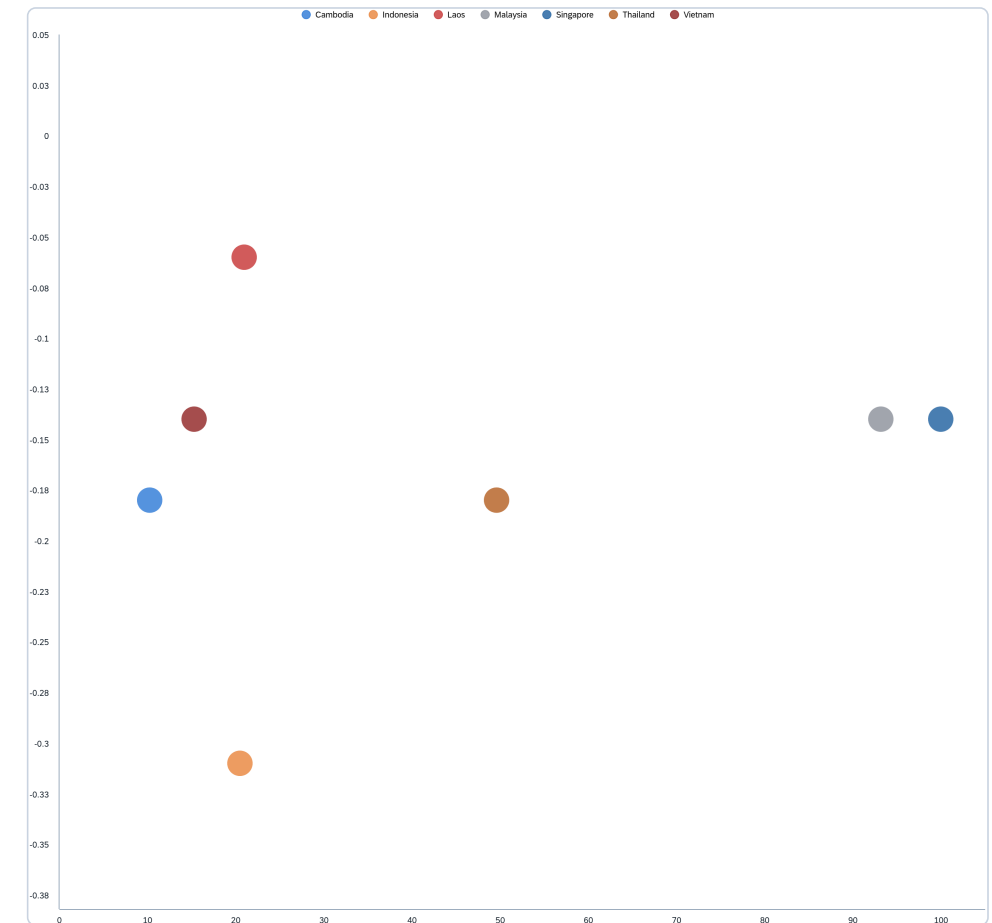
Sources: cost per case from peer-reviewed Indonesian studies; case counts from our own panel. The cheapest anchor (US\$90/episode) still gives US\$0.42M, so we have not chosen the most flattering figure.

## LIMITS &amp; DATA ETHICS

# What we tested and threw away

- **We falsified our own innovation angle.** “Drought  $\times$  water access” looked promising ( $r = -0.31$ ) but collapsed after controlling for El Niño (partial  $r = -0.15$ ) and showed no moderation by piped access ( $\rho = +0.34$ ,  $p = 0.46$ ,  $n = 7$ ). **We dropped the claim.**
  - **We do not claim a 2023–24 record**, because our panel is incomplete for Vietnam and the Philippines after 2022.
  - **Incidence only for 2018–2020**, the only years with official BPS denominators. Elsewhere we use raw counts and never rank regions by them.
  - **Flood and disaster records look correlated with dengue ( $r = +0.66$ ) until the counts are divided by population**, at which point the association reverses ( $r = -0.14$ ). We report the null.
  - Correlation is not causation. This is an **early-warning** claim, and the mechanism is cited from the literature.
- A rule that survives its own falsification tests is worth more to a health office than a larger claim that has never been tested.**

No relationship ( $\rho = +0.34$ ,  $p = 0.46$ ,  $n = 7$ )



THE PRODUCT

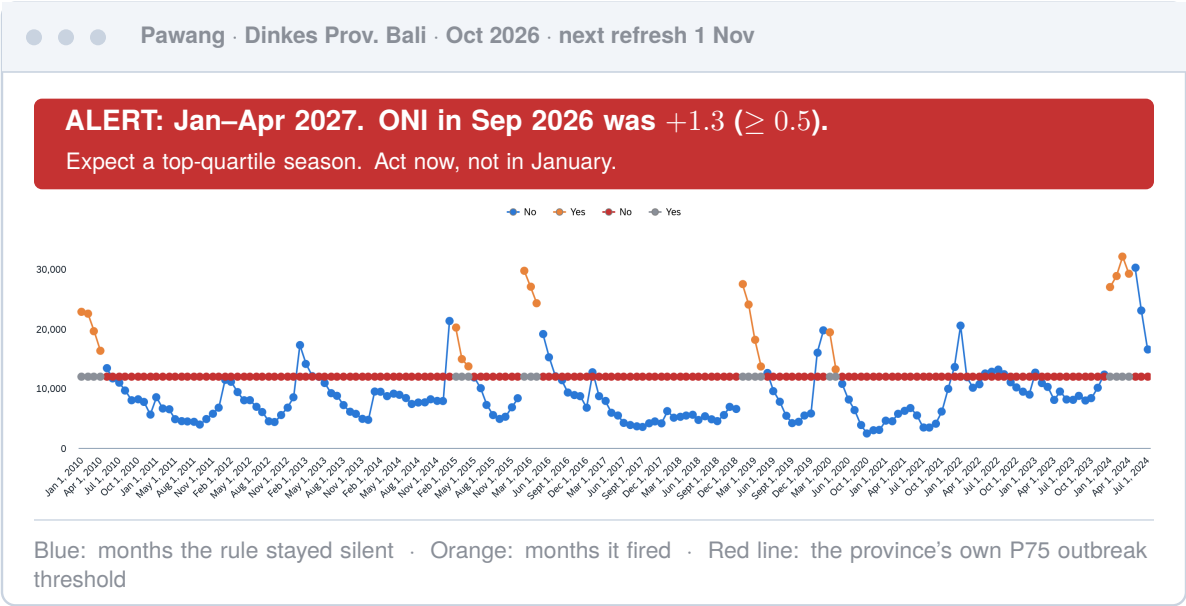
# Pawang: the rule, delivered on the first of every month

## Pawang

*Pawang* is the Indonesian and Malay word for someone who claims to know the weather months ahead. This one shows its working: a single SAC page that a district officer opens on the 1st, together with an automatic WhatsApp advisory. It escalates; it never declares an emergency.

The digital layer does the fetching, not the deciding. SAC refreshes the NOAA ONI feed and the office’s own case calendar on a schedule, and SAC Smart Predict runs beside the rule as a second opinion. The alert itself remains the two-condition rule, which can be audited by hand, so no one has to trust a model they cannot inspect.

### What Bali opens on 1 October 2026



### Who does what

| WHO                             | ROLE                                                                                                     |
|---------------------------------|----------------------------------------------------------------------------------------------------------|
| District health office gov’t    | Reads the advisory; moves PSN/3M, larviciding and stockpiles forward into the alerted window             |
| Ministry of Health · BMKG gov’t | Own the monthly refresh; BMKG’s forecast ONI stretches the notice past four months                       |
| Jumantik cadres NGO & community | Larval checks and household messaging concentrated in the alerted months instead of spread thin all year |
| Cloud & telco partners private  | Host the SAC tenant and the broadcast; one analyst-hour a month, no data team                            |

### SCALES BY SWAPPING ONE INPUT

Thailand already validates with its own predictor (temperature anomaly, lag 2, lift 2.7). Any ASEAN member repeats the fit on its own records.

## REFERENCES 1/2

# Data sources

## DATA

OpenDengue V1.3. doi:10.6084/m9.figshare.24259573.v4

World Bank Climate Change Knowledge Portal (ERA5 0.25° monthly temperature and precipitation).

NOAA Climate Prediction Center. Oceanic Niño Index (ONI v5).

WHO/UNICEF Joint Monitoring Programme for Water Supply, Sanitation and Hygiene, 2025 update.

BPS-Statistics Indonesia. Provincial population, WebAPI.

European Commission JRC. Global Human Settlement Layer (GHS-UCDB R2019A).

SIPSN, Ministry of Environment and Forestry, Indonesia.

BNPB. Indonesian Disaster Data and Information (DIBI).

## METHOD REFERENCE

Hussain-Alkhateeb L, et al. (2017). *Early warning and response system (EWARS) for dengue outbreaks: operational guide*. WHO/TDR technical document — no DOI assigned; the peer-reviewed description is Hussain-Alkhateeb 2018 on the next page.

## CLIMATE AND DENGUE

Tian Y, et al. (2025). Rising dengue risk with increasing El Niño–Southern Oscillation amplitude and teleconnections. *Nat Commun*. doi:10.1038/s41467-025-63655-0

Djaafara B, et al. (2026). Dengue transmission heterogeneity across Indonesia's archipelago: climate-driven spatiotemporal patterns and policy implications. *PLOS Negl Trop Dis*. doi:10.1371/journal.pntd.0014135

Mokhtar S, et al. (2024). Global risk of dengue outbreaks and the impact of El Niño events. *Environ Res*. doi:10.1016/j.envres.2024.119830

Sutriyawan A, et al. (2025). Impact of climate change on dengue incidence: a systematic review of evidence from Southeast Asia. *J Popul Soc Stud*. doi:10.25133/jpssv342026.028

Lowe R, et al. (2021). Combined effects of hydrometeorological hazards and urbanisation on dengue risk in Brazil. *Lancet Planet Health*. doi:10.1016/S2542-5196(20)30292-8

Gibb R, et al. (2023). Interactions between climate change, urban infrastructure and mobility are driving dengue emergence in Vietnam. *Nat Commun*. doi:10.1038/s41467-023-43954-0

Wang Y, et al. (2022). Impact of extreme weather on dengue fever infection in four Asian countries. *Environ Int*. doi:10.1016/j.envint.2022.107518

Choi Y, et al. (2016). Effects of weather factors on dengue fever incidence and implications for interventions in Cambodia. *BMC Public Health*. doi:10.1186/s12889-016-2923-2

Polrob P, et al. (2025). Nonlinear and lagged effects of climate variability on dengue incidence in Bangkok. *BMC Public Health*. doi:10.1186/s12889-025-25420-2

## REFERENCES 2/2

# Early-warning systems, cost, and validation

## EARLY-WARNING SYSTEMS

Hussain-Alkhateeb L, et al. (2018). Early warning and response system (EWARS) for dengue outbreaks: recent advancements towards widespread applications in critical settings. *PLOS ONE*.

doi:10.1371/journal.pone.0196811

Baharom M, et al. (2022). Dengue early warning system as outbreak prediction tool: a systematic review. *Risk Manag Healthc Policy*. doi:10.2147/RMHP.S361106

Benitez-Valladares D, et al. (2021). Validation of the early warning and response system (EWARS) for dengue outbreaks: evidence from the national vector control program in Mexico. *PLOS Negl Trop Dis*. doi:10.1371/journal.pntd.0009261

Cárdenas R, et al. (2022). The early warning and response system (EWARS-TDR) for dengue outbreaks: can it also be applied to chikungunya and Zika outbreak warning? *BMC Infect Dis*. doi:10.1186/s12879-022-07197-6

Schlesinger P, et al. (2024). Enabling countries to manage outbreaks: statistical, operational, and contextual analysis of the EWARS-csd for dengue outbreaks. *Front Public Health*. doi:10.3389/fpubh.2024.1323618

Quan TM, et al. (2026). Climatic determinants and simple thresholds for dengue early warning in Vietnam: a One Health perspective. *Sci One Health*. doi:10.1016/j.soh.2026.100159

## BURDEN, COST, AND VALIDATION

Nadjib M, et al. (2019). Economic burden of dengue in Indonesia. *PLOS Negl Trop Dis*. doi:10.1371/journal.pntd.0007038

Wilastonegoro NN, et al. (2020). Cost of dengue illness in Indonesia across hospital, ambulatory, and not medically attended settings. *Am J Trop Med Hyg*. doi:10.4269/ajtmh.19-0855

Shepard DS, et al. (2013). Economic and disease burden of dengue in Southeast Asia. *PLOS Negl Trop Dis*. doi:10.1371/journal.pntd.0002055

Shepard DS, et al. (2016). The global economic burden of dengue: a systematic analysis. *Lancet Infect Dis*. doi:10.1016/S1473-3099(16)00146-8

Sasmono RT, et al. (2026). Understanding the epidemiological trends and economic impact of dengue in Indonesia. *IJID Regions*. doi:10.1016/j.ijregi.2026.100860

Childs ML, et al. (2025). Climate warming is expanding dengue burden in the Americas and Asia. *PNAS*. doi:10.1073/pnas.2512350122